

Manipulator URDF Bundles Across Simulators: A Preliminary Study of Their Sim-to-Sim and Real-to-Sim Visual Matching

Yue Cao

Department of Electrical Engineering and Computer Science
Syracuse University, Syracuse, NY, USA
ycao70@syr.edu

Abstract

The simulator is an important tool in robot learning, since training directly on real robots is costly and risky. An increasing number of robot learning frameworks use camera views as a system input. This raises a question of whether different simulators render the same robot in the same way. This work focuses on a sub-area: the rendering of Unified Robot Description Format (URDF) bundles of manipulators. URDF is a widely used format for defining robot structure and kinematics. This preliminary study investigates two questions: *Q1 (Sim-to-Sim)*, do different simulators render the same manipulator URDF bundle in the same way, and *Q2 (Real-to-Sim)*, given a real robot, can its corresponding URDF bundle be identified from its appearance. To study these questions, this work first constructs a canonical-view rendering dataset of 131 manipulator URDF bundles across MuJoCo and PyBullet. For *Q1*, this work compares six visual similarity methods across all 131 manipulator URDF bundles in the dataset. These methods range from pixel-level and classical feature-based approaches to pretrained image embedding models, including CLIP, I-JEPA, DINOv2, and SigLIP2. For *Q2*, a preliminary real-to-sim retrieval demonstration is presented to explore whether a real robot photo can be matched to its corresponding URDF bundle.

Datasets — <https://github.com/BotYue/manipURDF4view>

Introduction

Simulation is widely used in robotics to reduce the cost and risk before deployment on real hardware. Its importance has grown further with the rise of robot learning, where models are often trained on simulated data before being transferred to physical systems (*Sim-to-Real*). A common robot description format used in simulation is the Unified Robot Description Format (URDF). It is in XML format and represents a robot as a tree of rigid links connected by joints. The visual and collision geometry of each link are specified using geometric primitives or references to external mesh files (e.g., STL, DAE). The URDF file together with its associated mesh files is referred to as a URDF bundle (Tola and Corke 2024).

Although URDF provides a shared representation of robot structure and geometry, the same URDF bundle can be interpreted and rendered differently across simulation environments. Differences in mesh handling, materials, camera configuration, lighting, and other rendering settings may change the visual appearance of the robot even when the underlying URDF bundle is unchanged. This is particularly relevant to robot learning systems that use cameras in the simulator as inputs. If robot appearance changes substantially across simulators, models and visual datasets developed in one simulator may not transfer directly to another. More broadly, this raises the question of whether different simulators render the same robot in the same way.

Therefore, this work considers two questions involving manipulator URDF bundles.

Q1 (Sim-to-Sim): do different simulators render the same manipulator URDF bundle in the same way?

Q2 (Real-to-Sim): given a real robot, is there a way to find its corresponding URDF bundle based on its appearance?

Q1 examines whether the visual identity of a URDF bundle is preserved across simulation environments. *Q2* extends this idea toward physical robots and asks whether visual correspondence could be established between a real robot and existing URDF assets. Such a capability could, for example, help identify an existing URDF bundle as a starting point when developing a representation for a physical robot. In this work, however, *Q2* is treated only as an exploratory direction rather than a quantitatively evaluated task.

The contributions of this work are summarized as follows:

- A canonical-view dataset of manipulator URDF bundles is rendered using MuJoCo and PyBullet.
- A sim-to-sim retrieval benchmark is developed to quantify visual correspondence across the two simulators using six visual similarity methods and single-view and multi-view settings.

Related Work

Cross-Simulator Comparison

Prior studies have compared robotics simulators primarily in terms of physics fidelity and computational performance. Bullet, Havok, MuJoCo, ODE, and PhysX were compared in robotic tasks with emphasis on simulation speed and physical accuracy (Erez, Tassa, and Todorov 2015). MuJoCo, Py-

Bullet, and V-Rep were evaluated with a Kinova manipulator, comparing simulated trajectories against real-world trajectories (Collins, Howard, and Leitner 2019). A broader review of physics simulators across a range of robots was presented in (Collins et al. 2021). Drake, MuJoCo, and Bullet were evaluated for impact and contact modeling on a bipedal robot (Acosta, Yang, and Posa 2022).

Visual Effects on Robot Learning

In the field of robot learning, recent studies have shown that learned policies can be sensitive to changes in visual input. In THE COLOSSEUM (Pumacay et al. 2024), five robot manipulation approaches, including R3M, MVP, PerAct, RVT, and VoxPoser, were evaluated under changes in object color, texture, lighting, background, and camera pose. These visual changes led to substantial differences in task success. In (Lou et al. 2026), Mask World Model, GE-ACT, and π_0 were evaluated under changes in background, lighting, and object color while task geometry was kept unchanged. These appearance shifts caused changes in manipulation success. In (Zhu et al. 2026), Vision-Language-Action models were evaluated under controlled changes in lighting and material properties. These rendering changes affected geometric reasoning and manipulation performance.

Canonical-View URDF Dataset

The dataset is derived from a public collection of URDF bundles developed by Tola and Corke (Tola and Corke 2024). This collection contains instances gathered from several repositories and software ecosystems. Only fixed-base robotic manipulators are retained in the dataset. The selected manipulators were rendered and validated in PyBullet and MuJoCo.

PyBullet natively supports URDF. However, MuJoCo does not use URDF as its native format. Instead, the URDF is converted into MuJoCo’s internal representation through the `MjModel.from_xml_string` interface.

Each manipulator is rendered from four canonical viewpoints: isometric, front, side, and top. These fixed viewpoints provide standardized observations across robots while capturing complementary aspects of their geometry. All manipulators are rendered using their default joint configuration.

In both simulators, the camera looks at the center of the robot’s bounding box, at a distance of $\max(2, 2d)$, where d is the bounding-box diagonal. The bounding box is computed differently: MuJoCo uses the robot’s visual mesh, while PyBullet uses its collision geometry. Camera orientation is set using an (azimuth, elevation) pair in MuJoCo and a (yaw, pitch) pair in PyBullet, so that each view matches the same physical viewpoint across simulators. This orientation setting is detailed in Table 1. PyBullet’s field of view is explicitly set to 35° to approximate MuJoCo’s framing, while MuJoCo’s is left at the renderer’s default. Finally, both simulators remove the background and set all non-robot pixels to white.

All images are rendered at a resolution of 1024×1024 pixels. Overall, 131 manipulators are successfully rendered

Table 1: Camera orientation for the canonical view.

View	MuJoCo (Az., El.)	PyBullet (Yaw, Pitch)
Isometric	$(45^\circ, -25^\circ)$	$(-45^\circ, -25^\circ)$
Front	$(90^\circ, 0^\circ)$	$(0^\circ, 0^\circ)$
Side	$(0^\circ, 0^\circ)$	$(-90^\circ, 0^\circ)$
Top	$(90^\circ, -89^\circ)$	$(0^\circ, -89^\circ)$

Table 2: Manufacturer and source repository composition of the canonical-view dataset ($N=131$).

Manufacturer	Count	Source Repository	Count
FANUC	42	ROS-Industrial	86
ABB	23	MATLAB Robotics	15
KUKA	19	Robotics Toolbox	13
Universal Robots	11	Drake	7
Stäubli	10	Others	10
Trossen Robotics	9		
Others	17		

in both simulators. Table 2 summarizes the composition of the dataset. The dataset is released as a GitHub repository <https://github.com/BotYue/manipURDF4view>.

Sim-to-Sim Matching

This study examines whether the same URDF bundle can be correctly matched across different simulation environments (*Q1, Sim-to-Sim*). Although the underlying URDF bundle is identical, differences in mesh interpretation, camera configuration, lighting, shadowing, and other renderer-specific settings can produce visible appearance differences. Representative examples are shown in Fig. 1.

In the matching test, MuJoCo renderings are used as queries and PyBullet renderings are used as the retrieval library. The evaluation is conducted across the full canonical-view dataset ($N=131$ manipulator URDF bundles), combining all URDF source repositories into a single retrieval pool.

Six methods are used to compare visual similarity. Two are classical image-based methods: pixel mean squared error (MSE) and Histogram of Oriented Gradients (HOG). The other four use pretrained image representations: CLIP (Radford et al. 2021), I-JEPA (Assran et al. 2023), DINOv2 (Oquab et al. 2024), and SigLIP2 (Tschannen et al. 2025).

For all four embedding-based methods, each image is encoded into a single feature vector using a pretrained model, and cross-simulator similarity is computed as the cosine similarity between vectors. We use the publicly released pretrained checkpoints, including: `openai/clip-vit-base-patch32`, `facebook/ijepa_vith16.1k`, and `google/siglip2-base-patch16-224` via Hugging Face, and `dinov2_vits14` via the official DINOv2 PyTorch Hub.

For each manipulator URDF bundle, similarity is computed for each of the four corresponding canonical views: isometric, front, side, and top. Two matching settings are evaluated. The first uses only the isometric view as a single-

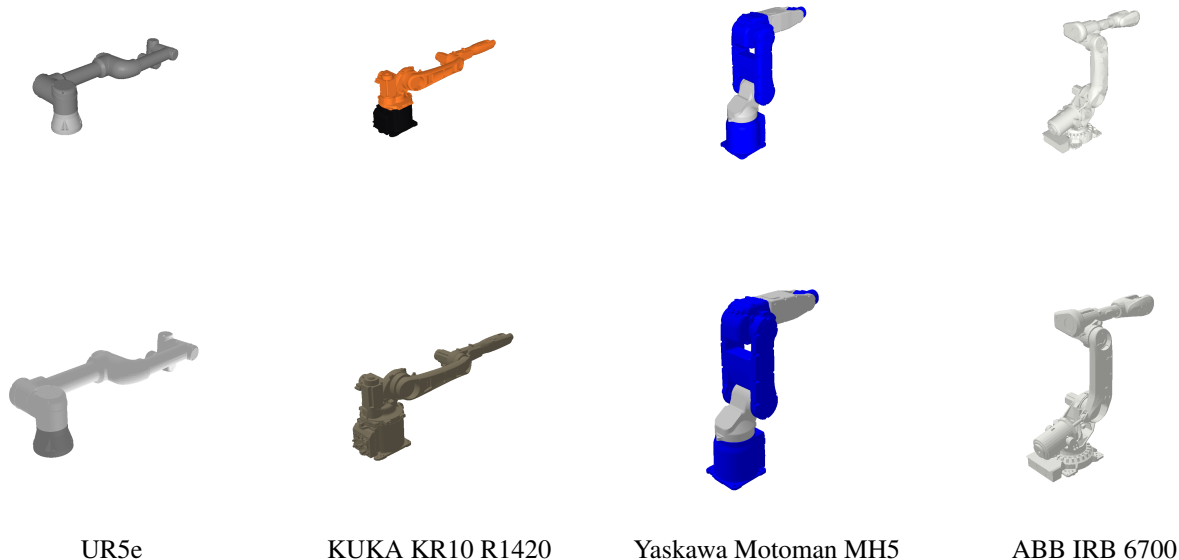


Figure 1: Isometric renderings of the same URDF bundle in MuJoCo (top row) and PyBullet (bottom row), for example manipulators.

view baseline, since it captures the robot structure along multiple spatial axes in a single image. The second averages the similarity scores from all four views with equal weight to test whether complementary viewpoints improve cross-simulator matching. For each MuJoCo query, all 131 PyBullet candidates are ranked according to the resulting similarity score.

The matching performance is evaluated using Top-1 and Top-5 accuracy. Top-1 matching accuracy is the percentage of queries for which the correct PyBullet bundle receives the highest similarity score. Top-5 accuracy is the percentage for which it appears among the five highest-ranked candidates.

The results are shown in Table 3. DINOv2 achieves the strongest matching accuracy in both the iso-only and four-view settings. SigLIP2 and CLIP both perform reasonably well, particularly at Top-5. For most methods, combining all four canonical views substantially improves accuracy over using the isometric view alone.

Future work will extend the sim-to-sim study to other commonly used simulation and visualization environments that support URDF (Tola and Corke 2023), such as Gazebo, RViz, and CoppeliaSim.

Real-to-Sim Matching

Q2 extends the same retrieval formulation from sim-to-sim matching to a real-to-sim setting. Instead of using a rendering from one simulator as the query, a photograph of a physical manipulator is used as the query, while simulated URDF renderings form the retrieval library.

Fig. 2 illustrates this preliminary pipeline. A real robot photograph is used as the query, and the retrieval library consists of MuJoCo isometric-view renderings of the ma-

Table 3: MuJoCo $\rightarrow$ PyBullet matching results across the full canonical-view dataset ($N=131$), comparing single-view (isometric only) similarity against four-view equal-weighted similarity.

Method	Iso-Only		4-View	
	Top-1	Top-5	Top-1	Top-5
Pixel MSE	7.6%	18.3%	8.4%	18.3%
HOG	9.2%	43.5%	17.6%	48.1%
CLIP	22.9%	50.4%	36.6%	63.4%
I-JEPA	29.8%	67.2%	38.2%	77.1%
DINOv2	42.0%	79.4%	61.8%	93.1%
SigLIP2	23.7%	61.8%	49.6%	90.1%

nipulator bundles in the dataset. Both the real photograph and every simulated rendering are encoded using the same pretrained DINOv2 model. Cosine similarity between their embeddings is then used to rank the candidate URDF bundles.

This real-to-sim component is currently an exploratory demonstration, and no quantitative real-to-sim evaluation is reported in this study. Compared with the controlled sim-to-sim setting, real-to-sim matching introduces additional variation from background, illumination, camera properties, viewpoint, and robot configuration.

Joint configuration is a particularly important limitation. The canonical-view dataset intentionally uses the default joint configuration for each robot so that Q1 can focus on differences between rendering pipelines. A physical robot, however, may appear in an arbitrary configuration. There-

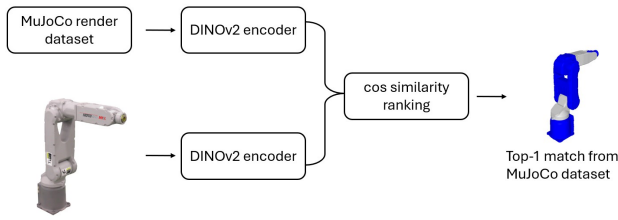


Figure 2: Real-to-sim matching pipeline. Both the query photo and every image in the MuJoCo rendering library are encoded by the same DINOv2 model into a shared embedding space.

fore, a more complete real-to-sim benchmark would require rendering each URDF bundle under multiple joint configurations, as well as collecting a larger set of corresponding physical robot images.

A potential future application of real-to-sim retrieval is URDF asset development. Given an image of a physical manipulator, visual retrieval could help identify an existing URDF bundle with similar appearance and structure that may serve as a starting point for further development. This application is not evaluated in the present study, but motivates further investigation of real-to-sim visual correspondence.

Conclusion

This preliminary study shows that the same manipulator URDF bundle can appear differently across MuJoCo and PyBullet, while pretrained image embeddings can still recover its identity across simulators. Multi-view matching generally improves performance over a single view. The real-to-sim component remains exploratory, and future work will extend the study to additional simulators and robot configurations.

References

- Acosta, B.; Yang, W.; and Posa, M. 2022. Validating robotics simulators on real-world impacts. *IEEE Robotics and Automation Letters*, 7(3): 6471–6478.
- Assran, M.; Duval, Q.; Misra, I.; Bojanowski, P.; Vincent, P.; Rabbat, M.; LeCun, Y.; and Ballas, N. 2023. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15619–15629.
- Collins, J.; Chand, S.; Vanderkop, A.; and Howard, D. 2021. A review of physics simulators for robotic applications. *IEEE Access*, 9: 51416–51431.
- Collins, J.; Howard, D.; and Leitner, J. 2019. Quantifying the reality gap in robotic manipulation tasks. In *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 6706–6712.
- Erez, T.; Tassa, Y.; and Todorov, E. 2015. Simulation tools for model-based robotics: Comparison of Bullet, Havok, Mujoco, Ode and Physx. In *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 4397–4404.

Lou, Y.; Chi, X.; Zhang, X.; Qian, Z.; Li, C.; Zhang, R.; Lyu, Y.; Song, G.; Fu, C.; Xu, H.; et al. 2026. Mask World Model: Predicting What Matters for Robust Robot Policy Learning. *arXiv preprint arXiv:2604.19683*.

Oquab, M.; et al. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.

Pumacay, W.; Singh, I.; Duan, J.; Krishna, R.; Thomason, J.; and Fox, D. 2024. THE COLOSSEUM: A Benchmark for Evaluating Generalization for Robotic Manipulation. In *Proc. Robotics: Science and Systems (RSS)*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 8748–8763.

Tola, D.; and Corke, P. 2023. Understanding URDF: A Survey Based on User Experience. In *Proc. IEEE Int. Conf. Automation Science and Engineering (CASE)*.

Tola, D.; and Corke, P. 2024. Understanding URDF: A Dataset and Analysis. *IEEE Robotics and Automation Letters*, 9(5): 4479–4486.

Tschannen, M.; et al. 2025. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. *arXiv preprint arXiv:2502.14786*.

Zhu, Y.; Wang, Z.; Yang, J.; Xie, J.; Yu, J.; Gu, J.; and Wang, B. 2026. Toward Visually Realistic Simulation: A Benchmark for Evaluating Robot Manipulation in Simulation. *arXiv preprint arXiv:2605.06311*.